in animals can be studied with reference to shell coiling in *Limnaea* and kappa particles in *Paramecium*.

4.11 Eugenics, Euphenics And Euthenics

Eugenics

Application of the laws of genetics for the improvement of human race is called **eugenics**. The term **eugenics** means "well born" and was coined by **Francis Galton** in 1885. For the betterment of future generations it is necessary to increase the population of outstanding people and to decrease the population of abnormal and defective people by applying the principles of eugenics.

Two methods of Eugenics

(i) Constructive method or Positive eugenics

(ii) Restrictive method or Negative eugenics

(i) Positive eugenics

Positive eugenics attempts to increase consistently better or desirable germplasm and to preserve the best germplasm of the society. The desirable traits can be increased by adopting the following measures:

(i) Early marriage of those having desirable traits

(ii) Subsiding the fit and establishing sperm and egg banks of precious germplasm

(iii) Educating the basic principles of genetics and eugenics

(iv) Improvement of environmental conditions

(v) Promotion of genetic research

(ii) Negative eugenics

Negative Eugenics attempts to eliminate the defective germplasm of the society by adopting the following measures:

(i) Sexual separation of the defectives

(ii) Sterilization of the defectives

(iii) Control of immigration and

(iv) Regulation of marriages

Euphenics

The symptomatic treatment of genetic disease of man is called Euphenics or Medical engineering. In 1960 Joshua Lederberg coined the term Euphenics. It means normal appearing. It deals with the control of several inherited human diseases especially the inborn errors of metabolism. Eg. Phenylketonuria (PKU)

Euthenics

The science of improvement of existing human race by improving the environmental conditions is called euthenics. It can be achieved by subjecting them to better nutrition, better unpolluted ecological conditions, better education and sufficient medical facilities.

Summary

Genetics is a branch of biology that deals with the study of heredity and variation. It describes how characteristics and features pass on from the parents to their offsprings in successive generations. Variation is the degree by which progeny differ from their parents. A set of three or more alleles of the same gene occupying the same locus in a given pair of homologous chromosomes controlling a particular trait is called Multiple allele. ABO blood grouping in man is a good example for multiple allelism. Apart from A and B antigens, the RBC's of humans contain a special type of antigen called Rh antigen/Rh factors. Erythroblastosis foetalis, also called haemolytic disease of the newborn, in which the red blood cells of a foetus are destroyed due to maternal immune reaction resulting from a blood group incompatibility between the foetus and the mother.

The mechanism of determination of male and female individuals in a species is called sex determination. The chromosomes are different in two sexes and referred to as allosomes; the remaining chromosomes are named autosomes. The inheritance of a trait that is determined by a gene located on one of the sex chromosomes

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